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Photocopy Master Sheets

Years 7-8

Astronomy

Disk filename = “09.Astronomy”

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Topics Available

Year 7-8 General Science

Disk Filename	Topic Name
01.Energy	Energy
02.Forces	Forces
03.Matter	Solids, Liquids & Gases
04.Mixtures	Separating Mixtures
05.Elements	Elements & Compounds
06.Cells	Living Cells
07.Life	Living Things
08.LifeSystems	Plant & Animal Systems
09.Astronomy	Astronomy
10.Earth	The Earth
11.Ecosystems	Ecosystems

Year 9-10 General Science

Disk Filename	Topic Name
12.Waves	Wave Energy (inc. Light)
13.Motion	Forces & Motion
14.Electricity	Electricity
15.Atoms	Atoms & Elements
16.Reactions	Compounds & Reactions
17.DNA	Cell Division & DNA
18.Evolution	Evolution of Life
19.Health	Health & Reproduction
20.Universe	The Universe
21.EarthScience	Earth Science
22.Resources	Resources & Technology

Year 11-12 Science Courses

Biology
Preliminary Core
Local Ecosystem
Patterns in Nature
Life on Earth
Evolution Aust. Biota
HSC Core
Maintain. a Balance
Blueprint of Life
Search for Better Health
Options
Communication
Genetics:Code Broken?

Chemistry
Preliminary Core
Chemical Earth
Metals
Water
Energy
HSC Core
Production of Materials
Acidic Environment
Chem.Monit.&Mngment
Options
Shipwrecks, Corrosion...
Industrial Chemistry

Earth & Envir. Science
Preliminary Core
Planet Earth...
Local Environment
Water Issues
Dynamic Earth
HSC Core
Tectonic Impacts
Environs thru Time
Caring for the Country
Option
Introduced Species

Physics
Preliminary Core
World Communicates
Electrical Energy...
Moving About
Cosmic Engine
HSC Core
Space
Motors & Generators
Ideas to Implementation
Options
Quanta to Quarks
Astrophysics

All Topics Available as PHOTOCOPY MASTERS and/or KCiC

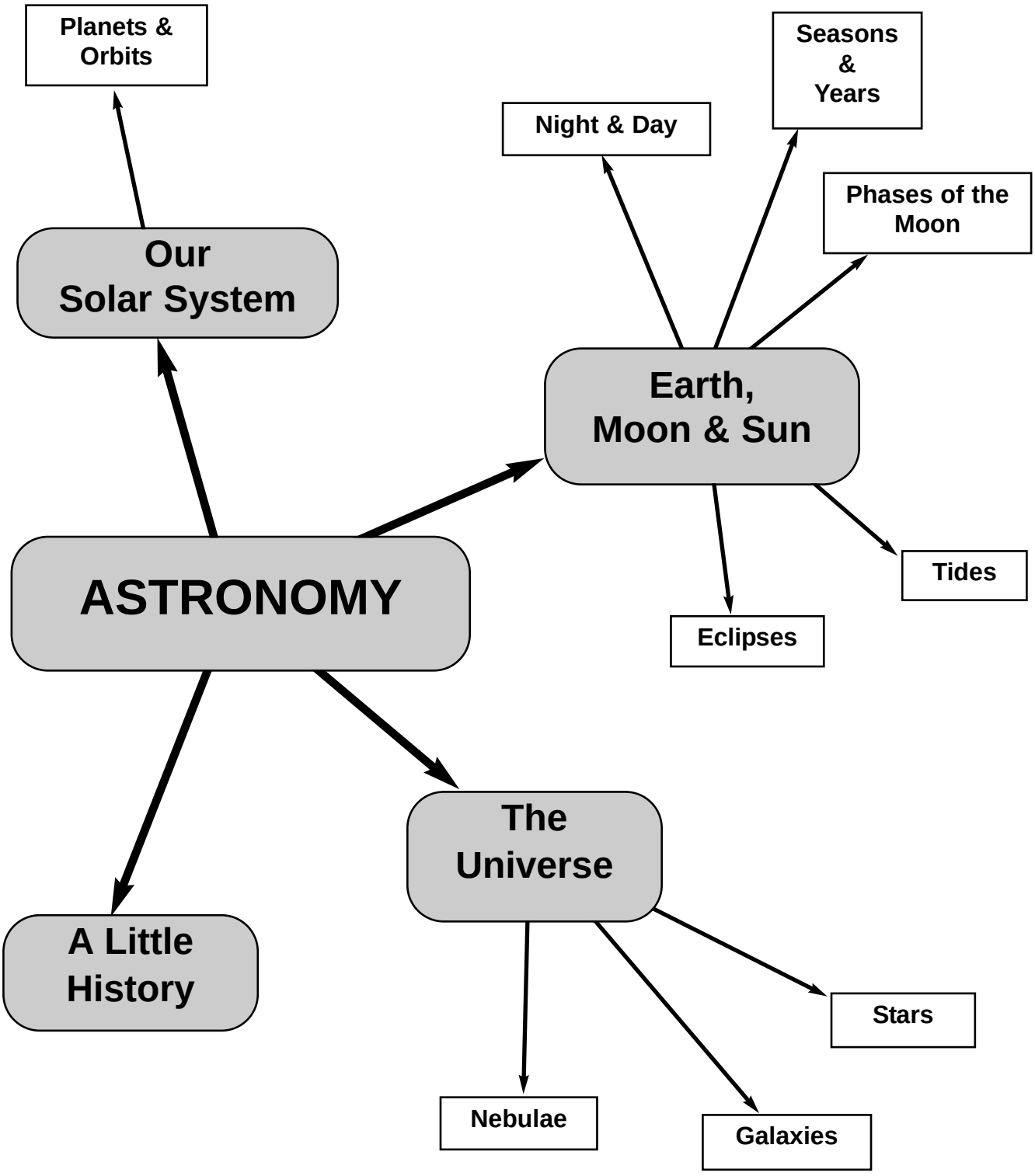
Photocopy Masters (PDF files)
Black & White, A4 portrait-orientation for clear, economical photocopying.

KCiC = Key Concepts in Colour
Full colour, formatted for on-screen study and data projection. PDF + Powerpoint®
Powerpoint is a trademark of Microsoft Corp.

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"Mind-Map" Outline of Topic

This topic belongs to the branch of Science called "Astronomy", the study of the Universe beyond the Earth. In this topic you will study our "neighbourhood" and the effects of the Sun and Moon on the Earth, as well as get an overview of the Universe beyond.



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Make your own "Mind-Map" TITLE PAGE.
Cut out the boxes. Sort them into an appropriate lay-out on a page of your workbook, then glue them down. Add connecting arrows and colour in.

ASTRONOMY

Our Solar System

Earth, Moon & Sun

A Little History

The Universe

Planets & Orbits

Phases of the Moon

Seasons & Years

Tides

Eclipses

Stars

Night & Day

Galaxies

Nebulae

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The Science of Astronomy

"Astronomy" comes from a Greek word which means literally "to look at the stars". The modern science of Astronomy includes the study of everything beyond the Earth. In practice, this usually ends up divided into 2 distinct areas of study: study of our Solar System, and study of the Universe beyond.

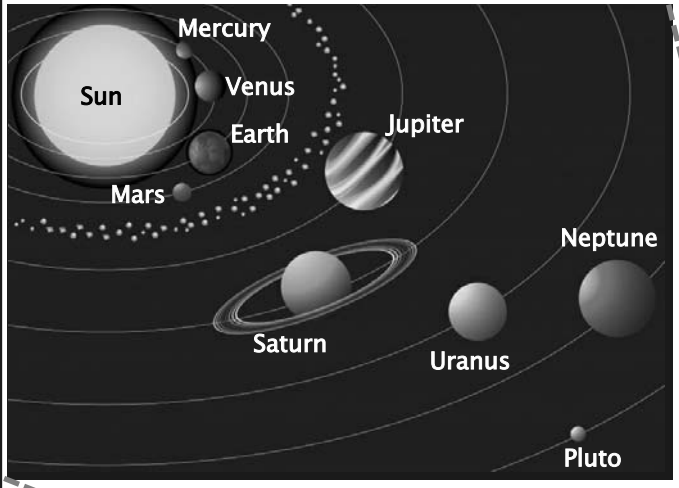
The Solar System

("solar" = Sun)

Our Sun is a star. It is a huge ball of hot, glowing gases.

In orbit around the Sun are 8 major planets, several minor planets and many smaller bodies such as asteroids and comets. The gravity of the Sun holds all these objects in orbit around it.

The Earth is one of the major planets, although certainly not the largest.



Most of the major planets also have moons. A moon is a minor planet which orbits around another planet, rather than directly around the Sun.

Our moon, the Moon, is one of the largest in the system. Together, the Earth-Moon pair are often considered a "double planet".

Until recently, the planet Pluto was considered one of 9 planets. Pluto has now been re-classified as a "minor planet", along with a few other medium-sized members of the system.

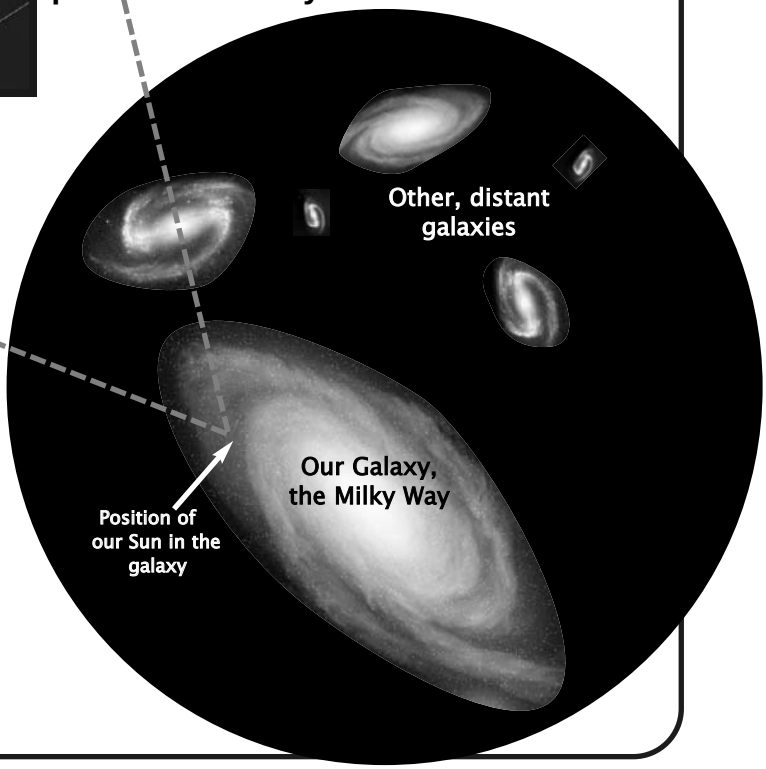
The Universe

Beyond our Solar System are billions of other suns. The stars are just like our Sun, but much further away, so they appear very small. Many (perhaps most of them) have their own solar systems of orbiting planets.

Stars are gathered together in huge swirling clumps called "galaxies". Our Sun is just one of over 200 billion stars in our galaxy, the "Milky Way".

The stars (and their solar systems) within a galaxy all swirl around each other in orbit. The gravity of all the material in the galaxy holds everything together.

Beyond our galaxy there are billions of other galaxies. The distances involved, and the total number of stars and planets is literally astronomical!



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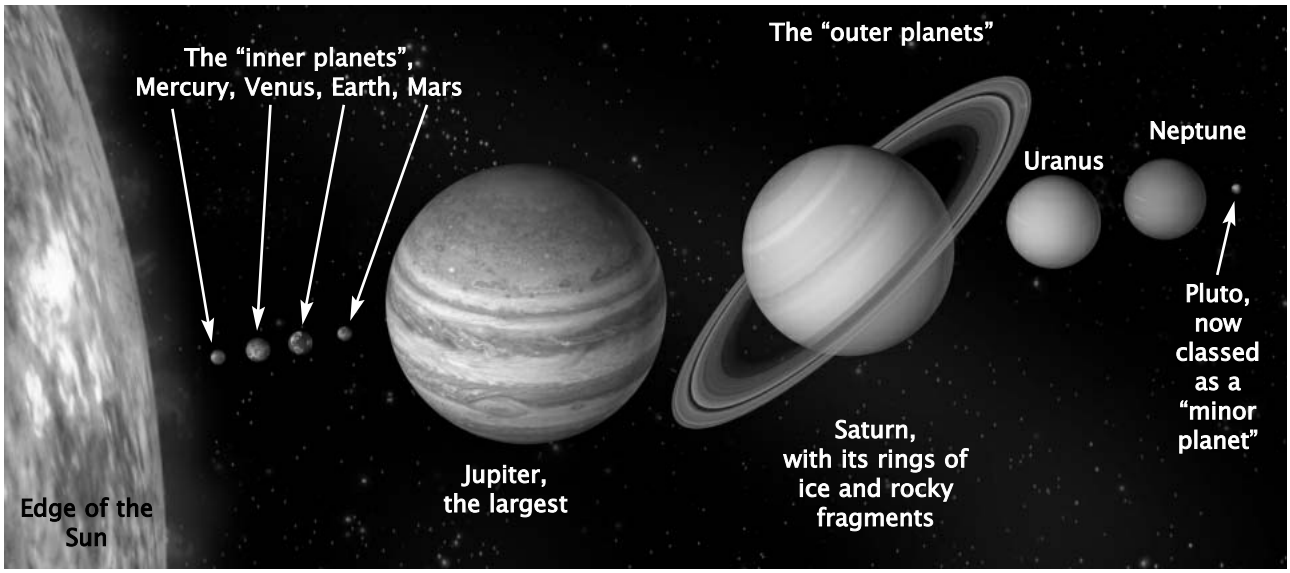
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The Members of the Solar System

It's not easy to get a clear understanding of the sizes of things, and the distances within our solar system... but we're going to try.

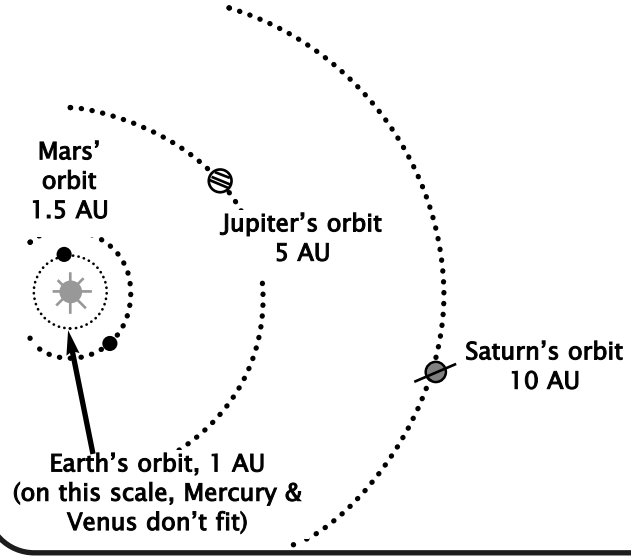
This diagram shows the relative <u>sizes</u> of the Sun and major planets.
The sizes are in proportion, but the distance between them is wildly wrong!



Planetary Orbits

The planets are in orbit around the Sun. The Sun's immense gravity is pulling them in, but their "sideways" speed prevents them falling into the Sun. The result is that they swing around the Sun in an orbit that is almost circular. With no friction in the vacuum of space, the orbiting can continue for billions of years.

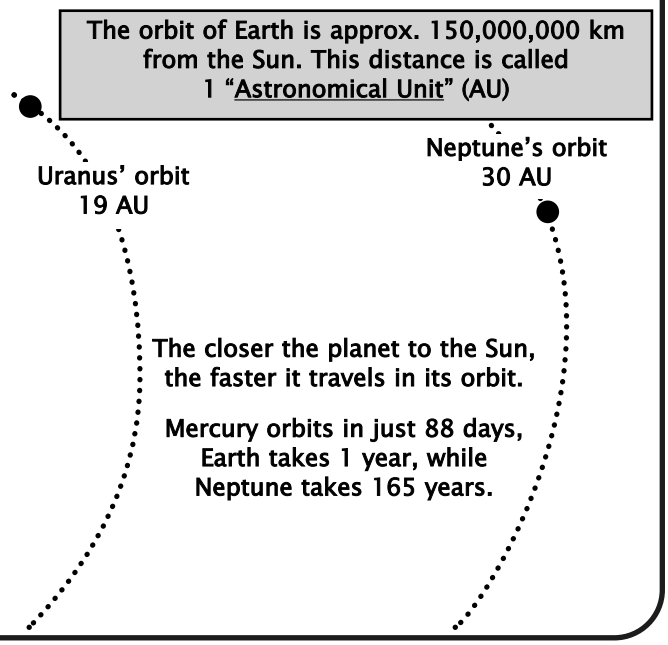
All the orbits lie roughly in the same flat plane around the Sun.



Orbital Distances

The diagram below is an attempt to show the orbits of the planets approximately to scale. This time, the sizes of the planets are wildly wrong.

Imagine you are out in distant space "above" the plane of planet orbits. The Sun would be like a very bright star, but the planets themselves would not actually be visible without a telecope.



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Worksheet 1

The Solar System

Student Name.....

1. Fill in the blank spaces

Our "solar system" comprises the Sun, which is a a)....., and all the b)..... and other objects which c)..... around it.

In order from the Sun the 8 major planets are d)....., e)..... f)....., g)....., h)..... i)....., j)..... and k).....

Most planets also have 1 or more l)..... which orbit around them.

An orbit is controlled by the force of m)..... This pulls the planet (or moon) inwards, but it cannot simply fall down because it has a lot of "sideways" n).....

2. Answer the questions.
(for some you may need research)

a) Name the largest planet.

b) Which planet has a prominent system of rings around it?

c) Which 2 planets are the only ones without at least 1 moon?

d) In between the orbits of which 2 planets is the zone called the "asteroid belt"?

e) Which planet was recently re-classified as a "minor planet".

Worksheet 2

A Model of the Solar System

It is very difficult to make a model of the Solar System which shows both sizes and distances to the same scale. This exercise is an attempt to do that in a simple, rough way.

Approx. scale used is 1m = 1 million km

Scale Models of Sun & Planets.
Sun: use a large beachball,
or simply a garbage bin.

Planets
Earth needs to be about 1cm in size.
Use a marble, bead or simply a button.
Venus: same size bead/button as Earth.
Jupiter... about 10cm... use a softball.
Mercury... about 3mm... a rice grain.

1. Place the garbage bin "Sun" at the edge of a sporting field area.

2. Pace out or measure 60m. Place a rice grain here (on a sheet of paper so you don't lose it) to represent planet Mercury.

3. Go a further 40m and place a small marble to represent Venus.

4. Move out another 50m to place the "Earth" marble or bead.

This is about all you can manage on most sports fields.

5. If there's room, pace out another 600m to place the softball (or grapefruit) Jupiter model.

In case you're wondering, to place another garbage bin to represent the nearest star to our solar system, you would need to move twice around the world to stay in scale.

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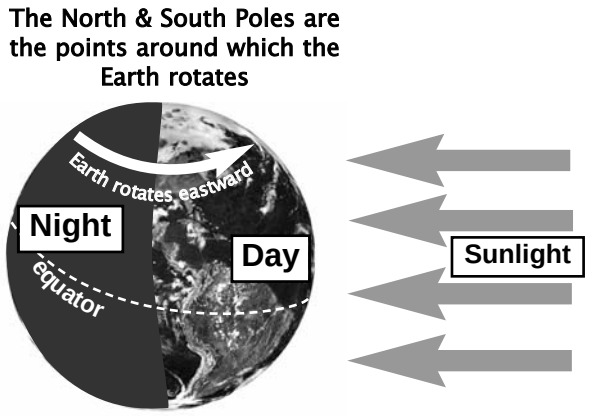
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Earth & Sun

**The most obvious and important object we see in the sky is the Sun.
From Earth, it appears to move across the sky from east to west.
The Sun rising and setting gives us night and day.
We have a cycle of seasons which gives us our "year".
These are the result of the motion of the Earth as it spins
on its axis and orbits around the Sun.**

Night & Day

Although it looks like the Sun moves across the sky each day, it is really the Earth spinning like a top which gives us night and day.



Since the Earth spins around once every 24 hours, the cycle of light and dark repeats every 24 hours... one day.

The Earth rotates towards the east. That's why we see the Sun appear from the eastern horizon at dawn, and "set" in the west.

The rotation of the Earth defines our most basic time unit... the day. Other planets rotate at different speeds, so their "day" can be quite different.

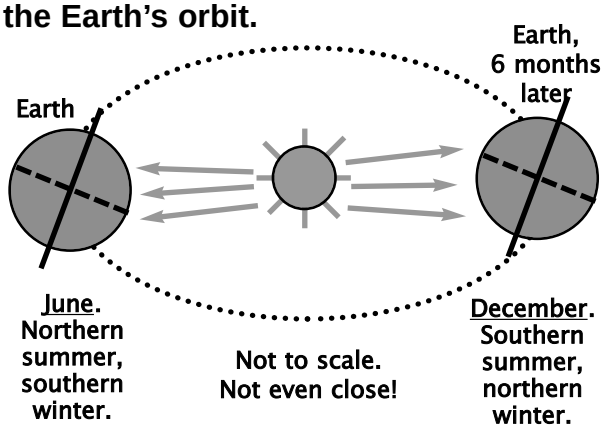
Jupiter, for example, spins very fast and a "Jupiter day" is only about 10 hours long. Venus spins so slowly that a "Venusian day" lasts about 8 months!

We don't always get 12 hours light and 12 hours dark. That is controlled by how far from the equator you are located, and by the seasons... read on.

Seasons & Years

What causes summer and winter?

The axis of rotation of the Earth (which passes through the North & South Poles) is tilted, compared to the path of the Earth's orbit.



In June, the tilt of the Earth's axis causes the northern hemisphere to receive more light and heat, and the south less.

Six months later, the Earth has moved to the other end of the orbit, but it's still tilted the same way, so the seasons are reversed.

In between, in March and again in September, each hemisphere receives about the same amount of heat and light.

The time it takes to get back to the starting point and complete the cycle is what we call a "year".

It actually takes $365^{1}/_{4}$ days. We have 365 days in a calendar year, then add an extra day every 4th year. Without "leap years" our calendar would get out-of-step with the seasons.

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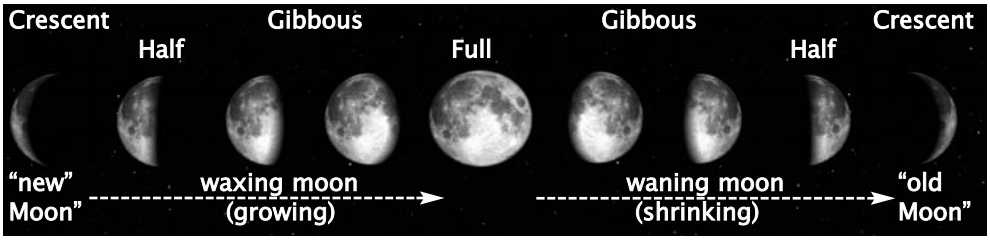
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Earth & Moon

Just like the Sun, the Moon also appears to rise in the east, travel across the sky, and set in the west. This daily (or nightly) movement across the sky is due to the same effect as the moving Sun; it is the <u>rotation of the Earth</u> which makes the Moon appear to travel across the sky from "rising" to "setting".

However, every day the Moon rises about an hour later than the day before, and it appears to change its shape over a cycle that takes about a month.

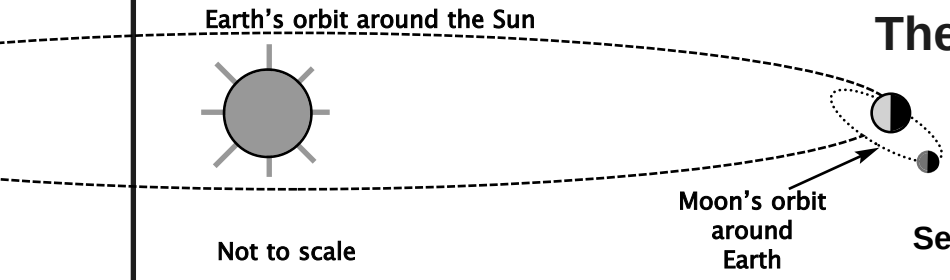


The Phases of the Moon

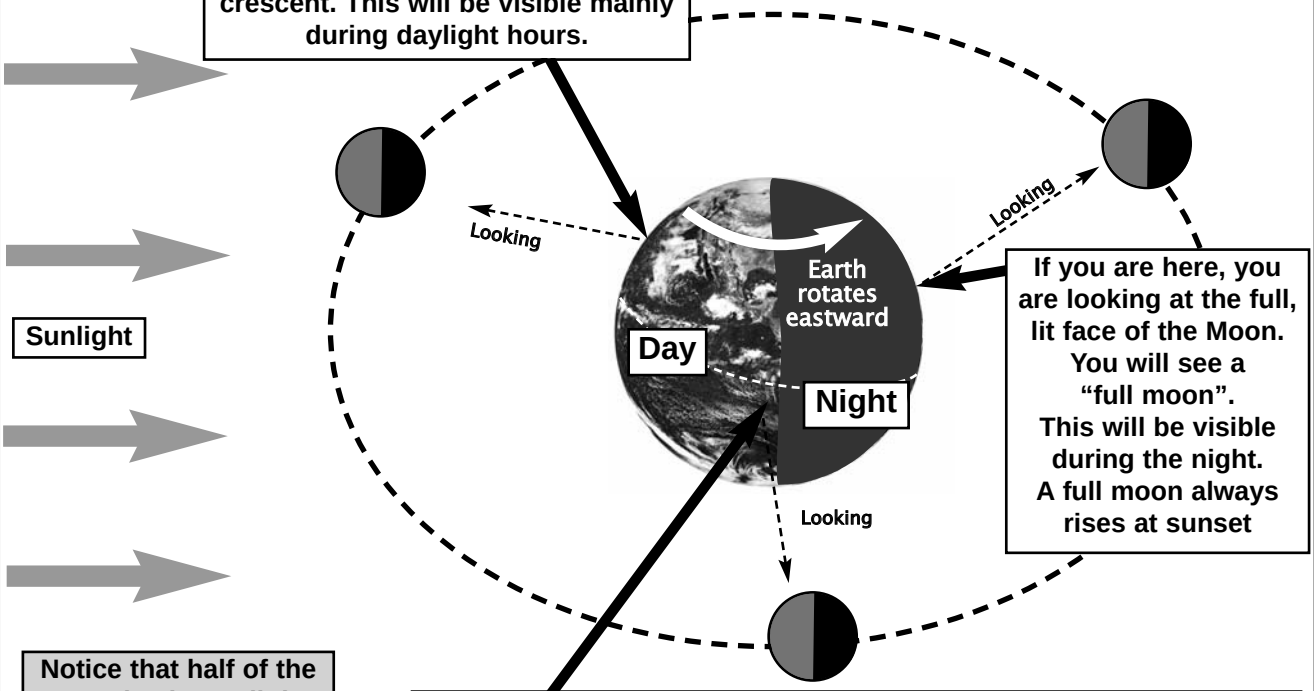
Firstly, you must realise that we see the Moon entirely by the <u>sunlight</u> which reflects from it.

Secondly, you must know that the Moon is a satellite of the Earth; it is in orbit around the Earth, and a full orbit takes about $29^1/_2$ days.

The rest is geometry.



If you are here on Earth looking up at the Moon, you are looking mainly at the dark, un-lit part of the Moon. All you will see is a thin, lit edge or crescent. This will be visible mainly during daylight hours.



If you are here, you are looking at the full, lit face of the Moon. You will see a "full moon". This will be visible during the night. A full moon always rises at sunset

Notice that half of the Moon is always lit by the Sun, but we can only see <u>part</u> of the lit face most of the time.

If you are here, you are looking at the Moon "side-on" and will see a "half- moon". This will be visible partly by day and by night. A half-moon rises about midday or midnight.

Your teacher might use models to demonstrate this in 3-dimentions.

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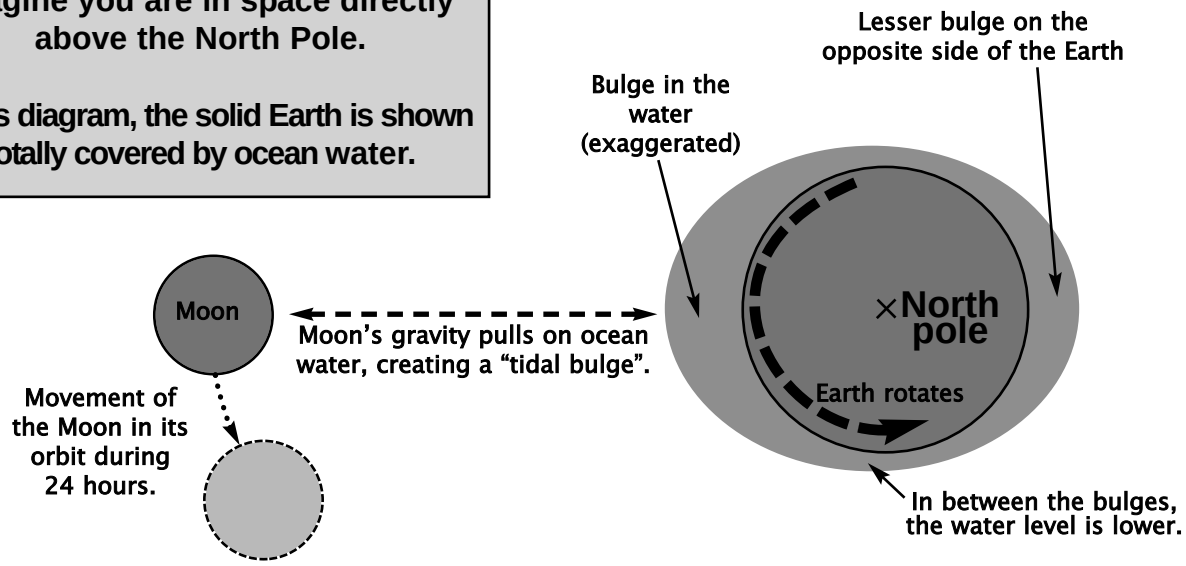
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Sun, Moon & the Tides

If you live near the coast, you know that the water level at the beach rises and falls every day... high and low tides come and go, roughly every 6 hours. The tides are caused mainly because of the gravity of the Moon tugging on the liquid oceans on Earth. There is also a lesser effect from the gravity of the Sun.

Imagine you are in space directly above the North Pole.

In this diagram, the solid Earth is shown totally covered by ocean water.



Think about what happens in each 24 hour period.

The Moon only moves a little way along its orbit and the "bulge" of water that its gravity attracts stays underneath it.

However, the Earth itself has spun around a complete 360º rotation.

This means that each part of the Earth rotates through 2 tidal bulges every day.

The water level in a bulge is slightly higher and we see the water creep up the beach to a high tide, twice per day.

Each part of the Earth also rotates through 2 low tides, where the water level is lower between the bulges.

Actually, we don't quite see a complete set of tides in 24 hours. Because the Moon moves along its orbit, it actually takes about 25 hours to go through a cycle of 2 high tides and 2 low tides.

Effect of the Sun on the Tides

Although the Sun is much larger than the Moon, and its gravity much stronger, it is also much further away. The Sun's gravity does have an effect on the tides, but it is much less than the effect of the Moon.



Sometimes the gravity of the Sun and Moon both pull in the same line. This creates bigger tidal bulges, so the high tides are higher, and low tides are lower. These are called "spring tides".
This occurs when the Moon is in crescent phase, and again at full moon.

At half-moon phases the Moon's gravity pulls at right angles to the Sun's gravity. This tends to make tidal bulges smaller. High tides are lower and low tides are higher. These lesser tides are called "neap tides".



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Worksheet 3

Sun, Earth & Moon

Here is a list of facts about the movements of the Earth & Moon.

Answer the questions which follow the list.

A. The Earth orbits around the Sun.

B. The Earth spins on its axis.

C. The Moon orbits around the Earth.

D. The Earth's axis of rotation is tilted compared to the plane of its orbit.

E. We see only those parts of the Moon which are lit by sunlight.

Student Name.....

Questions

1. Which unit of our time is defined by:

a) movement A? .....

b) movement B? .....

c) movement C? .....

2. Which one of the movements A-E causes day and night on Earth? .....

3. Which two of the items A-E cause the changing phases of the Moon? ..... and .....

4. Which two of the items A-E cause the seasons on Earth?

..... and .....

5. Which one of the movements A-E causes the Sun to move across the sky? .....

Worksheet 4

Sun, Earth & Moon

Fill in the blank spaces

It looks like the Sun moves across the sky each day. This is actually caused by the Earth a)..... on an axis through the north & south b)..... It takes c)..... hours for one complete rotation.

The Earth also d)..... around the Sun once every e)..... $^1/_4$ days, which we call one f)..... On our calendar we save up the $^1/_4$ days, and add 1 extra day each "g)..... year".

The Earth's axis of h)..... is tilted compared to the plane of the i)..... around the Sun. This causes the j)..... One hemisphere of the Earth receives more k)..... and ..... (energy) than the other, and then l)..... months later this is reversed.

Student Name.....

The movement of the Moon across the sky each day or night is also caused by m).....

The Moon seems to change its shape in a cycle which lasts about n)..... $^1/_2$ days. This is because the Moon o)..... the Earth, and because we can only see the Moon when p)..... As it changes positions in its orbit, we can see different amounts of q)..... ....., so it seems to change shape.

The tides are caused by the r)..... of the s)..... pulling on the ocean waters so that they "t)....." upwards. As the Earth rotates, the coastlines pass through a bulge (this is a u)..... tide) and then pass through a v)..... tide between the bulges.

The Sun's w)..... also affects the x)..... of the tides, but it has less effect than the Moon.

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Worksheet 5

Graphing Skills

Phases of the Moon

Student Name.....

The data in the following tables shows the time of "Moon-rise" every 2nd day over a one month period. (Times have been rounded to the nearest half-hour for simplicity.)

Graph this data in the grid below using a series of dots. When finished, connect the dots to form a line graph.

Times of Moonrise

Day	Time of Moonrise
2	12:30am
4	2:00am
6	4:00am
8	5:30am
10	7:00am

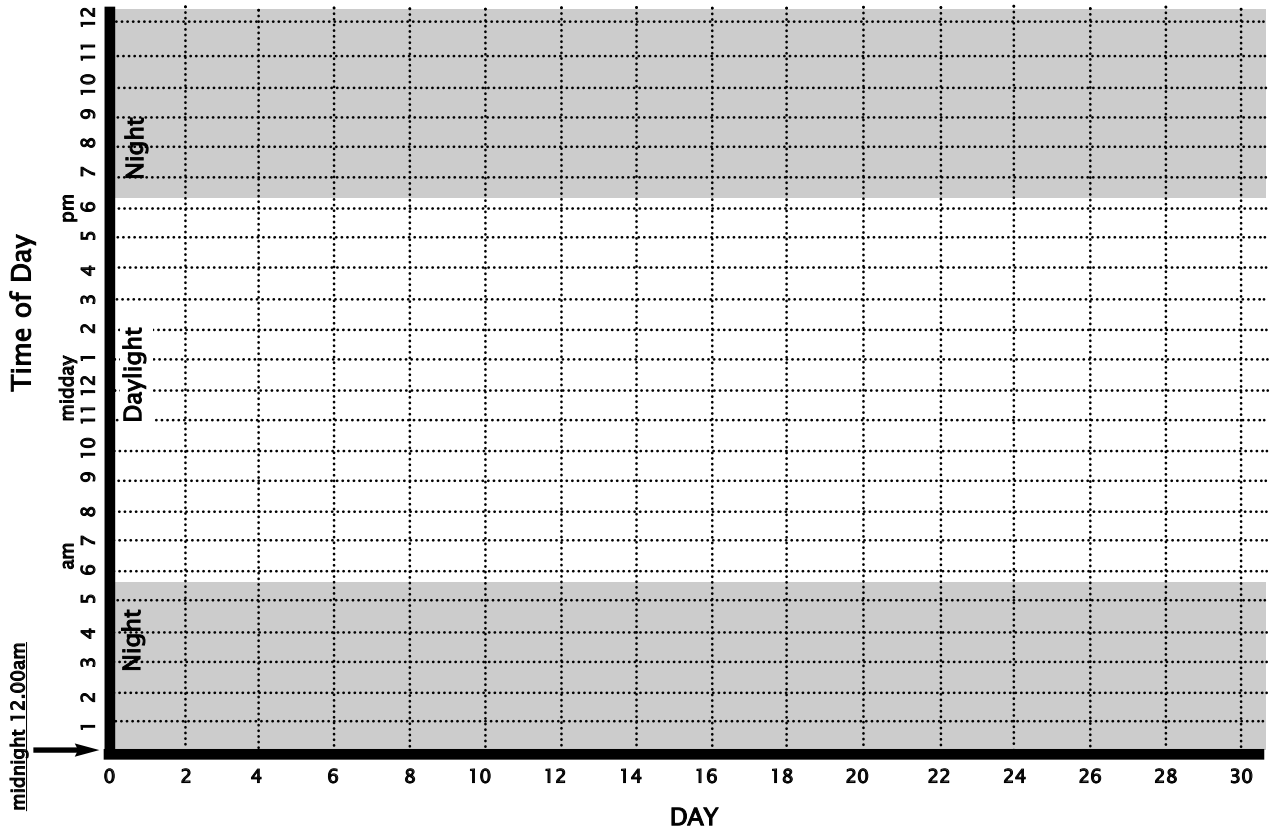
Day	Time of Moonrise
12	9:00am
14	10:30am
16	12:00pm
18	2:00pm
20	3:30pm

Day	Time of Moonrise
22	5:00pm
24	7:00pm
26	8:30pm
28	10:00pm
30	11:30pm

The Moon phases within this time were as follows:

Phase	Symbol	Day
half-moon (waning)	◗	1
old moon	)	7
new moon	(	8
half-moon (waxing)	◖	16
full moon	●	23
half-moon (waning)	◗	30

When you are finished the graph, sketch the symbol for each phase on the graph in the appropriate position.



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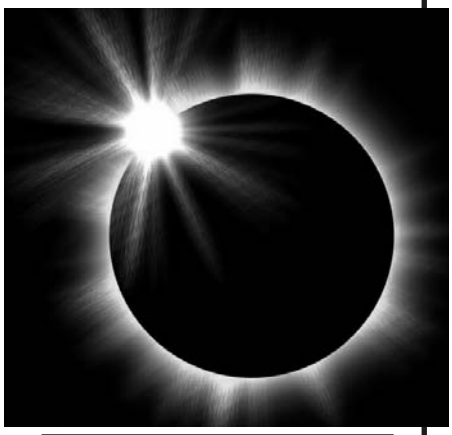
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Solar Eclipses

(solar = Sun, eclipse = to blot out)

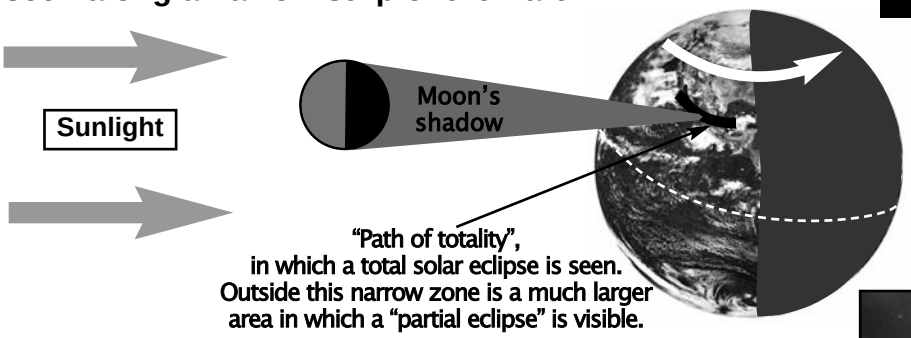
A solar eclipse occurs when the Moon gets exactly in-between the Sun and Earth, and the Moon's shadow sweeps across part of the Earth.



You don't see the Moon because from Earth you are looking at the completely un-lit side of the Moon. What you see is the entire face of the Sun blotted out, with only the glowing "corona" visible. It becomes dark in the middle of the day, and the stars appear in the dark sky.

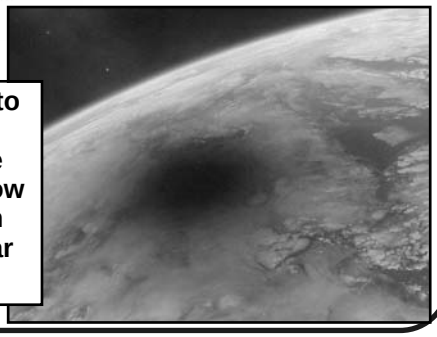
No wonder that ancient people were terrified!

A total eclipse lasts only a few minutes, and can only be seen along a narrow strip of the Earth.



Solar eclipses happen somewhere in the world almost every year. However, they can be seen in such a narrow area that you might wait 100's of years for one to occur where you are.

Think about the geometry of Sun, Moon and Earth during an eclipse and you might predict that the tides would be extreme. You'd be correct... high high tides and low low tides

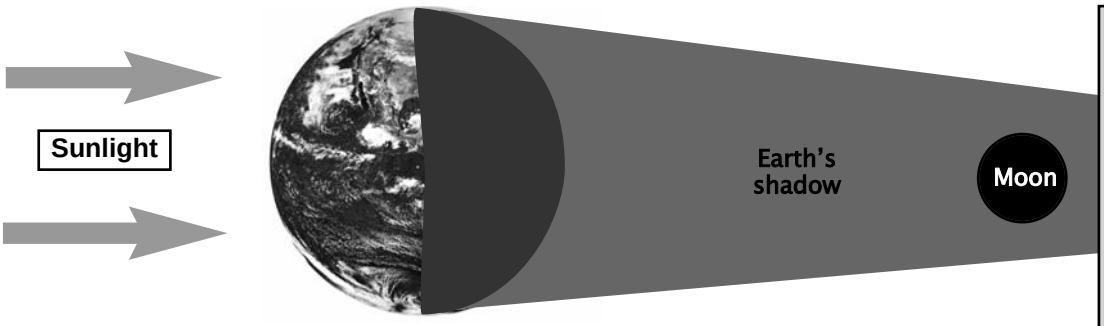


This is a photo taken from space of the Moon's shadow on the Earth during a solar eclipse

Lunar Eclipses

(lunar = Moon)

A lunar eclipse occurs when the Moon goes through the Earth's shadow. The Moon seems to disappear because we only see it by reflected sunlight, so it becomes invisible when in shadow.



Lunar eclipses occur more often than solar eclipses and can be seen from half of the Earth. They last longer too; close to 2 hours.

If you think about the geometry, you'll realise that a lunar eclipse occurs only when the Moon is at full-moon phase.
To see a lunar eclipse you must be on the dark side of the Earth, so it can be seen only at night time.

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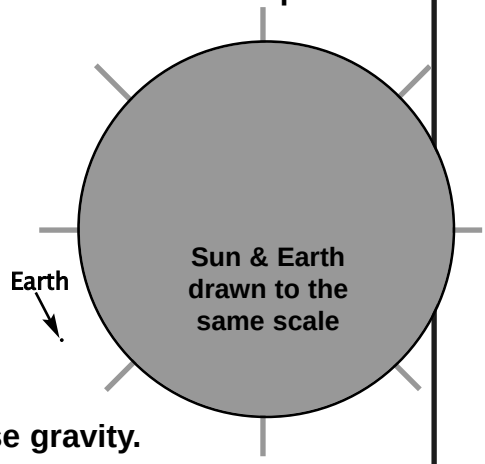
The Sun

Our Sun is a star. A rather small-sized, quite normal star.



Although it is 150 million km away we can feel its warmth so it must be very hot. In fact, we know that it is about 5,000^{o}C at the surface. Deep down inside its temperature is millions of degrees!

It's big too.
The Sun is over a million km in diameter.



The Sun is not solid...
no substance can be solid at such temperatures.

In fact, the Sun is a huge ball of hot glowing gas (mostly hydrogen & helium) held together by its immense gravity.

Energy in the Sun

The huge amount of heat and light energy coming from the Sun comes from nuclear reactions occurring deep inside.

The Sun is like a huge nuclear bomb.
It would explode, except it is so huge that its gravity holds the explosion together in a ball.
The energy comes to the surface and radiates off into space.

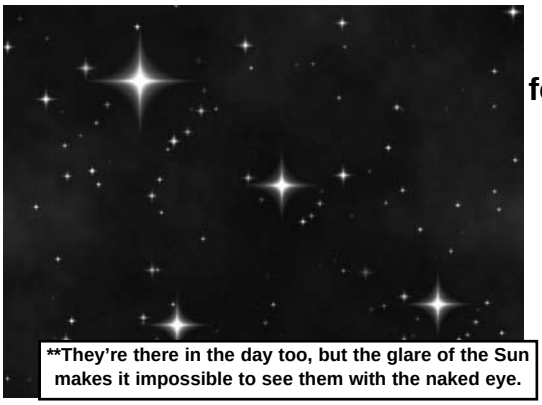
Stars & Planets in the Night Sky

On a clear night the sky is full of stars. Each star is another Sun similar to ours. Each one is a nuclear furnace burning furiously at millions of degrees. The stars appear very small because they are much, much further away.**

Distance to the Stars

The next nearest star to our solar system is over 30 million million km away. At the speed of the Space Shuttle it would take about 150,000 years to get there.

To measure such enormous distances, astronomers use the "light-year". This is the distance that a beam of light can travel in one year. Since light moves at 300,000 km/second, a light year is a very long way.



****They're there in the day too, but the glare of the Sun makes it impossible to see them with the naked eye.**

Spotting a Planet

Not all the stars in the night sky really are stars. A few are planets in our solar system, and therefore really quite close.

Planets do not produce light, but only reflect the light of the Sun. To spot them, look for bright "stars" that do NOT "twinkle".

Venus is the easiest to find. Look for a very bright, non-twinkling star low in the western sky just after sunset.

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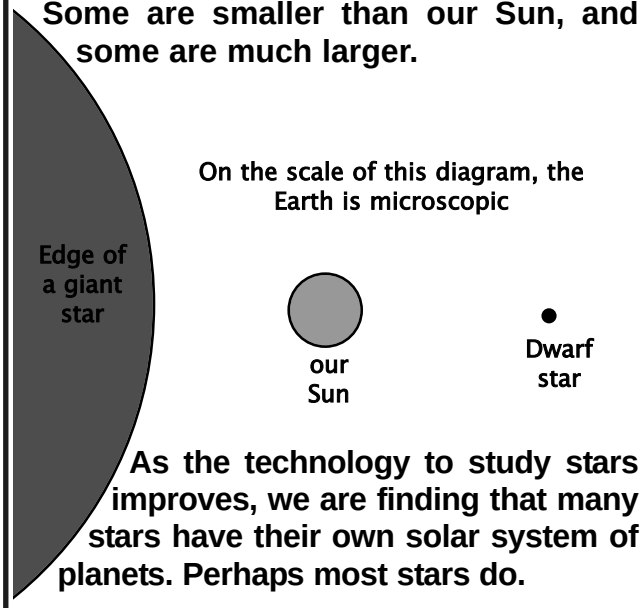
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Stars, Galaxies & Nebulae

Stars

Except for the few planets visible to the naked eye, all the points of light in the night sky are stars.

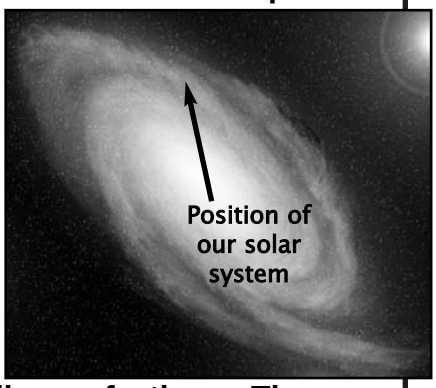
Each one is a huge gas ball of nuclear reactions held together by gravity. Some are smaller than our Sun, and some are much larger.



As the technology to study stars improves, we are finding that many stars have their own solar system of planets. Perhaps most stars do.

Galaxies

All of the easily visible stars in the night sky belong to our galaxy, the "Milky Way". It is a great swirling mass of billions of stars rotating in a gravitational orbit in a great flattened disk shape.



Our Sun and the solar system rotate right around the galaxy every few 100 million years.

Beyond our galaxy are billions of others. They are mostly millions of light years apart. This means that when we look at them (with a telescope) we are seeing light that left there millions or billions of years ago. That means we see them, not as they are now, but as they were millions of years ago... we are looking back in time!

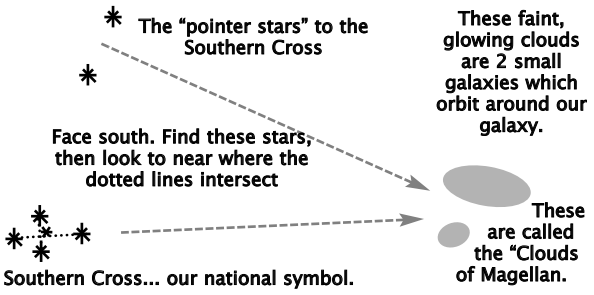
Nebulae

("Nebulae" is plural. The singular is "nebula" which means "a cloud" or "mist")
Before telescopes were invented, early astronomers had to rely on naked eye observations. They noticed that there were some objects in the night sky which were vague "blobs", or clouds. Some glowed with light, so they were called "bright nebulae", while others were dark, opaque clouds called "dark nebulae". With modern technology, we now know what they are:

Bright Nebulae

Bright, glowing nebulae are galaxies or clusters of stars which are so far away that our eyes cannot pick out individual stars. We see just a vague, blurry, glowing cloud.

Use this diagram to see the "Clouds of Magellan". Choose a clear, moon-less night, and get well away from streetlights.



Dark Nebulae

Dark nebulae actually are clouds. Made of gas and dust, they fill vast volumes of space within our galaxy. The dust is thought to be the "ashes" from an exploded giant star.



Eagle Nebula
NASA & ESA

These clouds are the places where new stars and solar systems are made. Our solar system is thought to have formed in a cloud like this, about 5 billion years ago.

Parts of the nebula in the photo are just beginning to glow from the new stars that have been "born" inside it.

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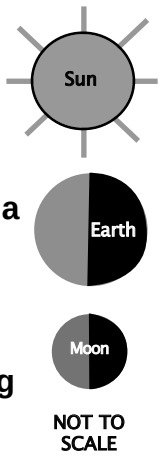
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Worksheet 6

Modelling Eclipses

1. <u>Solar Eclipse</u>



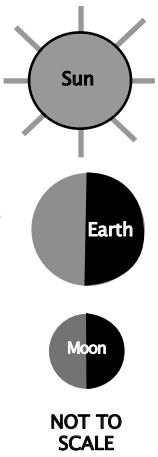
Cut out these models of Sun, Earth & moon.

Arrange them to show the relative positions during a solar eclipse.
Glue into final positions.

Sketch in any shadow which is important to understanding the solar eclipse.

Write a simple explanation for why the <u>Sun</u> cannot be seen during a total solar eclipse.
<u>and</u>
Write a simple explanation for why the <u>Moon</u> cannot be seen during a solar eclipse.

1. <u>Lunar Eclipse</u>



Cut out these models of Sun, Earth & moon.

Arrange them to show the relative positions during a lunar eclipse.
Glue into final positions.

Sketch in any shadow which is important to understanding the lunar eclipse.

Write a simple explanation for why the <u>Sun</u> cannot be seen during a lunar eclipse.
<u>and</u>
Write a simple explanation for why the <u>Moon</u> cannot be seen during a total lunar eclipse.

Worksheet 7

Sun, Stars, Galaxies, Nebulae

Fill in the blank spaces

Student Name.....

The Sun is a a)..... It is a huge ball of hot, glowing b)....., mainly c)..... and .....

Its energy comes from d)..... reactions, which would cause it to explode except its e)..... holds it together in a ball.

All the stars in the night sky are like the Sun, but they seem very small because they are f).....

Astronomers measure the distance to the stars in "g)..... - ....." This is the distance that h)..... can travel in i).....

Stars are grouped together in huge swirling groups called j)..... Our galaxy is called the "k)..... ....." and contains l)..... of stars. The stars in a galaxy are held together by m).....

Beyond our galaxy are n)..... of other galaxies. They are millions or billions of o).....-..... away.

"Nebula" means "p)....." Bright nebulae are q)..... or star clusters which look like clouds because we cannot see the individual stars.

Dark nebulae are clouds of r)..... and ....., possibly from an exploded star. New stars and their s)..... are formed inside these nebulae.

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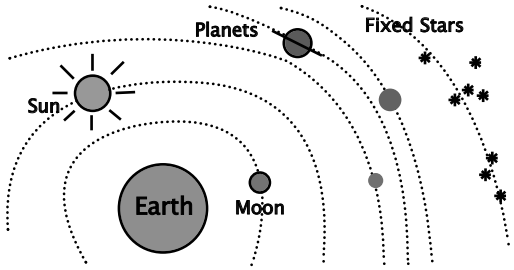
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A Little History

In ancient times, people worshipped the Sun as a god, or believed that the Sun, Moon and stars were the home of gods. (In modern language we still refer to the sky as "the heavens".) Starting about 2,500 years ago, some people began to try to understand the Universe in scientific terms. Astronomy is the oldest Science.

Earth-Centred or Sun-Centred?

For most of the history of Astronomy, people believed that the Sun, Moon, planets and stars all revolved around the Earth.



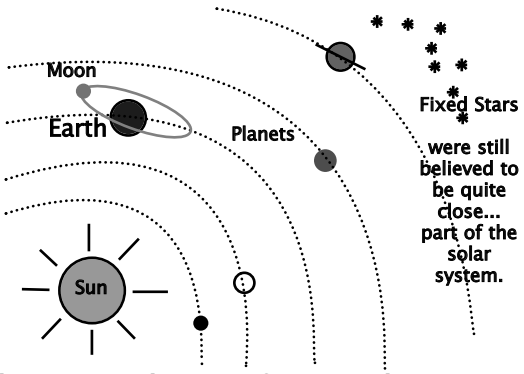
All these "heavenly bodies" were thought to be just a few kilometres up in the sky (or a few hundred km at most).

Some ancient Greeks had realised that you could explain things with the Sun at the centre. If the Earth went around the Sun, it also needed to rotate on its axis every day, so that the Sun would <u>appear</u> to move across the sky.

Ancient observers could find no evidence that the Earth was moving, so they concluded that the universe was Earth-centred.

Nicholas Copernicus (1473-1543)

Copernicus was the first person for over 1,500 years to seriously propose that the Sun was at the centre, and the Earth must orbit and spin.



This "new" theory of Copernicus was not immediately accepted. There was still no evidence for a moving Earth, and during 1,000 years or so, the Church had adopted the Earth-centred idea as a religious truth. People who believed in "revolution" could be punished!

Tycho Brahe (1546-1601)

Tycho used the most advanced observatory of the time to gather very accurate measurements of the movements of the planets. These were naked-eye measurements, but he used large mechanical devices to measure angles very accurately, and the best clocks to measure time.

He favoured the Earth-centred idea, and hoped his data would prove Copernicus wrong. When Tycho died, his student Kepler used his precious measurements.

Johannes Kepler (1571-1630)

Using Tycho's data, Kepler was able to develop a set of mathematical "laws" which described the orbiting of planets (including Earth) around the Sun. It all seemed to work, but there was no explanation for how planets revolved without falling down, and there was still no proof that the Earth moves.

Galileo Galilei (1564-1642)

Galileo was the first to use a <u>telescope</u> to observe the planets. He noticed that the planet Venus goes through "phases" like the Moon. This can only be explained by a Sun-centred model.

He also saw that Jupiter had moons of its own, which contradicted the "official" belief that everything must revolve around Earth.

Sir Isaac Newton (1642-1727)

Newton's <u>Theory of Gravity</u> provided the explanation for things to be "in orbit", and did away with the clumsy "crystal spheres" of previous models.

From his equations for Gravity, Newton could prove Kepler's Laws mathematically... this proved that the Sun-centred idea was correct.

Since the time of Newton, the Sun-centred model has been accepted as the scientifically correct description of the solar system, but it took another 200 years to discover the full story of stars, galaxies and distances... and we're still learning.

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New Technology Can Transform Science

The Telescope



The first primitive telescopes were invented in the 16th century. They had low magnification, but were soon very popular in the military for observing troop movements and enemy fortifications.

In trading cities like Venice, telescopes became vitally important for spotting approaching ships. To begin with, telescopes were not scientific tools.

It was Galileo who first used the new invention to view the night sky. What he saw revolutionised (literally... see below) the science of Astronomy.

The Earth is Not the Centre

Galileo saw that Jupiter had moons, but everything was supposed to orbit the Earth. He saw mountains and valleys on the Moon, but it was supposed to be a "perfect" heavenly body, not lumpy and messy like Earth. He saw that Venus went through "phases" like the Moon. That can only be explained if both Venus and Earth orbit around the Sun.

Galileo's telescope set Astronomy, and Science generally, off in a new direction... a direction that led straight to modern Science.



Society Affects Scientific Development

The Copernicus Model

Nicholas Copernicus knew his new idea was trouble, that's why he arranged for it to be published after his death. It was called "The Revolution of the Heavenly Spheres".

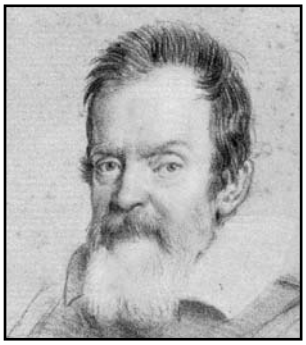
His idea was that the Earth, planets and stars revolved around the Sun. This was in direct conflict with the "official" view of the Catholic Church.

The Church saw any new idea as a threat to its authority, so Copernicus's book was banned, and anyone who supported the "revolution of the Earth" idea could be tortured and executed.

When Galileo made his discoveries by telescope he realised that Copernicus was correct. To avoid trouble with the Church, he published his findings in the form of a play, with ficticious characters debating and revealing the facts and theories about the idea of "revolution".

The word "revolution" took on a new meaning which we still use today... to bring in new ideas & new systems.

Galileo's Trial



Galileo's ploy did not work. He was arrested and threatened with torture. He was forced to publicly declare that Copernicus was wrong.

Finally, he was convicted and placed under house arrest for the rest of his life. (Anyone less important would probably have been burned alive... the usual punishment.)

This action basically stopped all further research and scientific development in Catholic countries. Scientists were too scared to do anything new, in case it was "wrong" in the eyes of the Church.

However, in Protestant countries like England and Holland, the new ideas were accepted and Science flourished.

The next generation of great scientists were English (e.g. Newton) or Dutch (Kepler) and they led a true scientific revolution.

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Topic Test
Astronomy

Student Name..... Score = /20

Answer all questions in the spaces provided.

1. (4 marks) (-1 for each error)
The major planets of the Solar System are: (in alphabetical order)

Earth, Jupiter, Mars, Mercury, Neptune, Saturn, Uranus, Venus.

Write them in order, outward from the Sun

....., ..., ...

....., ..., ...

....., ...

2. (6 marks)
a) What force controls the orbits of the planets? .....

b) What movement is the basis of each of these units of time?

i) day .....

ii) year .....

iii) month .....

c) Draw a simple sketch to show the positions of the Earth, Sun and Moon during a solar eclipse.

3. (10 marks)
Match each description to an item from the list. To answer, write the letter (A,B,C, etc) of the list item beside the description.

Description	matches with	**List Item**
i) The star at the centre of our solar system.		**.....**
ii) A blurry "cloud" among the stars.		**.....**
iii) Gravity from this mainly causes the tides.		**.....**
iv) The largest planet in our solar system.		**.....**
v) Many, many stars swirling around each other.		**.....**
vi) Type of eclipse when the Moon is in the Earth's shadow.		**.....**
vii) A measure of distance.		**.....**
viii) Time to orbit around the Sun.		**.....**
ix) Energy that powers the stars.		**.....**
x) An object that we see by reflected sunlight which changes shape.		**.....**

List Items Not all will be used.
Some may be used more than once.

A. lunar	**G. Galaxy**
B. nebula	**H. asteroid**
C. year	**I. nuclear**
D. light-year	**J. Jupiter**
E. Moon	**K. week**
F. Sun	**L. Pluto**

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Answer Section

Worksheet 1

1.

a) star	b) planets
c) orbit	d) Mercury
e) Venus	f) Earth
g) Mars	h) Jupiter
i) Saturn	j) Uranus
k) Neptune	l) moons
m) gravity	n) movement/speed

2.

a) Jupiter
b) Saturn
c) Mercury & Venus
d) Mars-Jupiter
e) Pluto

Worksheet 3

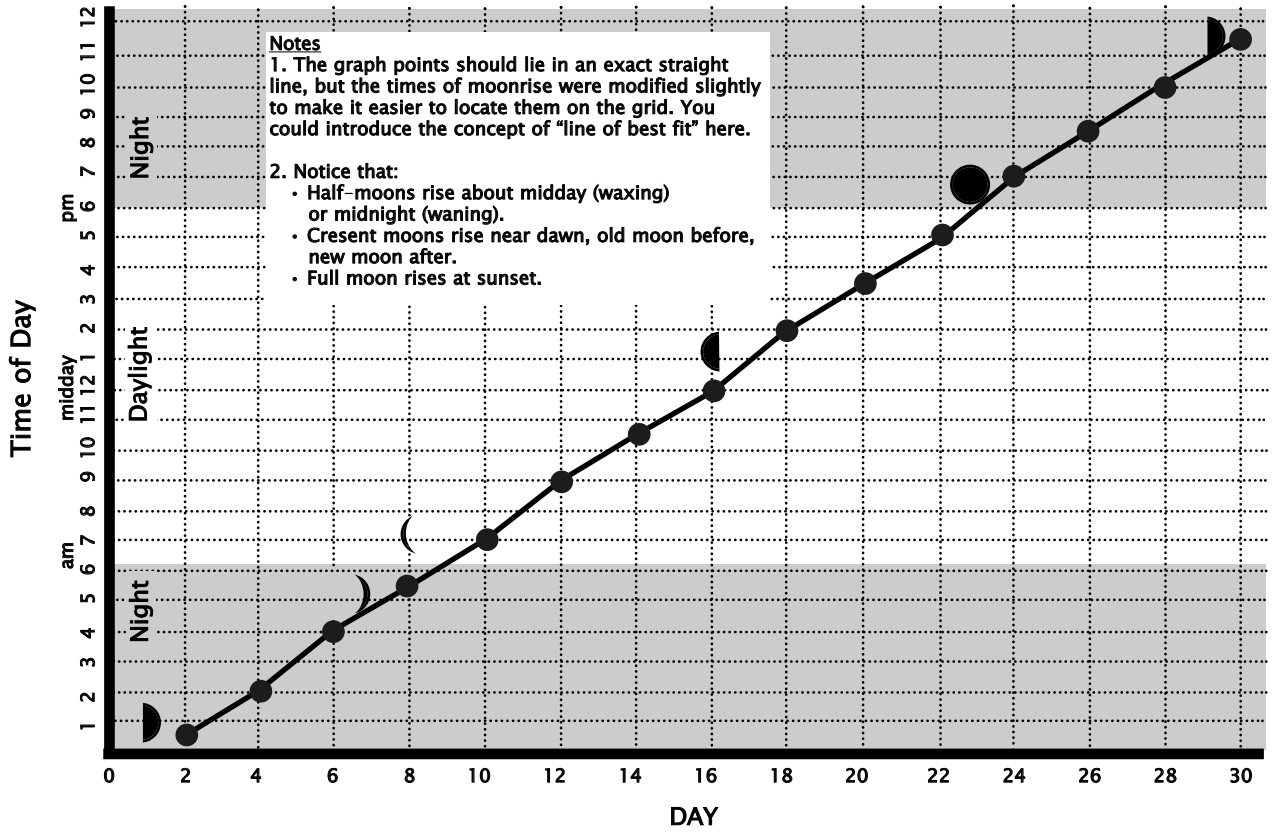
1. a) year b) day c) month
2. B
3. C & E
4. A & D
5. B

Worksheet 4

a) spinning or rotating

b) poles	c) 24
d) orbits	e) 365
f) year	g) leap
h) rotation	i) orbit
j) seasons	k) heat & light
l) 6	m) the Earth's rotation
n) 29	o) orbits
p) sunlight reflects from it	
q) the lit surface	r) gravity
s) Moon	t) bulge
u) high	v) low
w) gravity	x) height

Worksheet 5

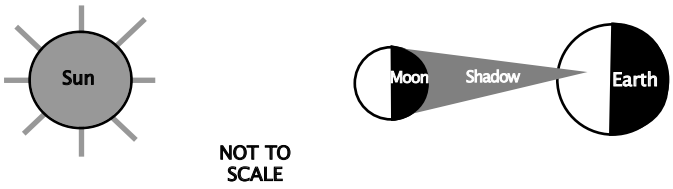


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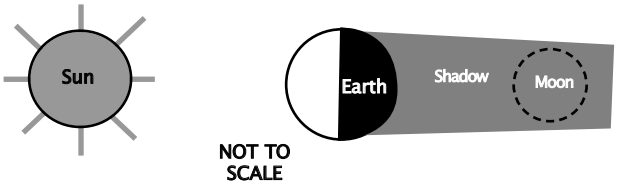
1. Solar Eclipse



Sun cannot be seen because the Moon completely covers it.

Moon cannot be seen because you are looking at the completely unlit, dark side of the Moon.

2. Lunar Eclipse



To see a lunar eclipse, you must be on the "night-side" of Earth, and therefore the Sun is not in sight.

We only see the Moon by reflected sunlight. In a total lunar eclipse, no sunlight reaches the Moon (it's in Earth shadow) so we cannot see it.

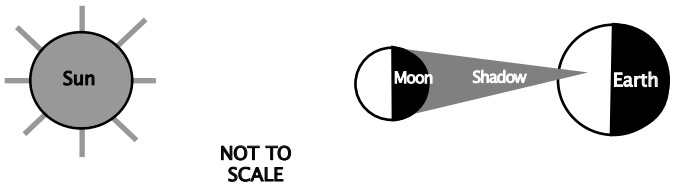
Worksheet 7

a) star **b) gases**
c) hydrogen & helium
d) nuclear **e) gravity**
f) so far away **g) light-years**
h) light **i) 1 year**
j) galaxies **k) Milky Way**
l) billions **m) gravity**
n) billions **o) light-years**
p) cloud **q) galaxies**
r) gas & dust **s) solar systems**

Topic Test

1.
Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune

2.
a) gravity
b)
i) Spinning (rotation) of the Earth.
ii) Orbit of Earth around the Sun.
iii) Orbit of Moon around Earth.
c)



3.

i) F	**vi) A**
ii) B	**vii) D**
iii) E	**viii) C**
iv) J	**ix) I**
v) G	**x) E**

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